

Iterators agents



```
state = {"goal": "Book PHL→YVR trip", "context": user_input}
while True:
    action = llm(f"State:{state}\nProvide action json or 'done'")
    if action == "done":
        break
    result = run_tool(action["tool"], **action["args"])
    state["context"] += result
```

Iteratively books flights & hotel, checking prices, updating state until “done.”

LLM **owns a loop**: plan $\rightarrow$ act $\rightarrow$ observe $\rightarrow$ repeat until done.

What LLMs

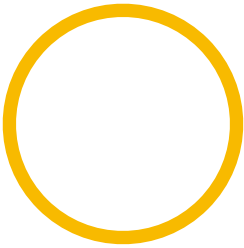


Routes

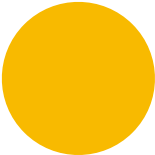
To allers

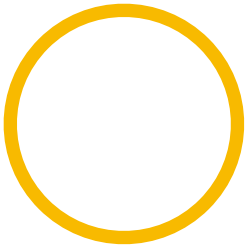
FullAutonomy



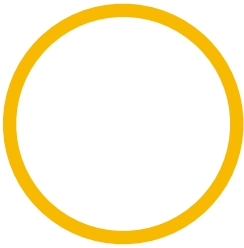




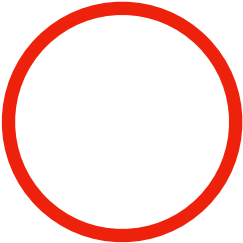




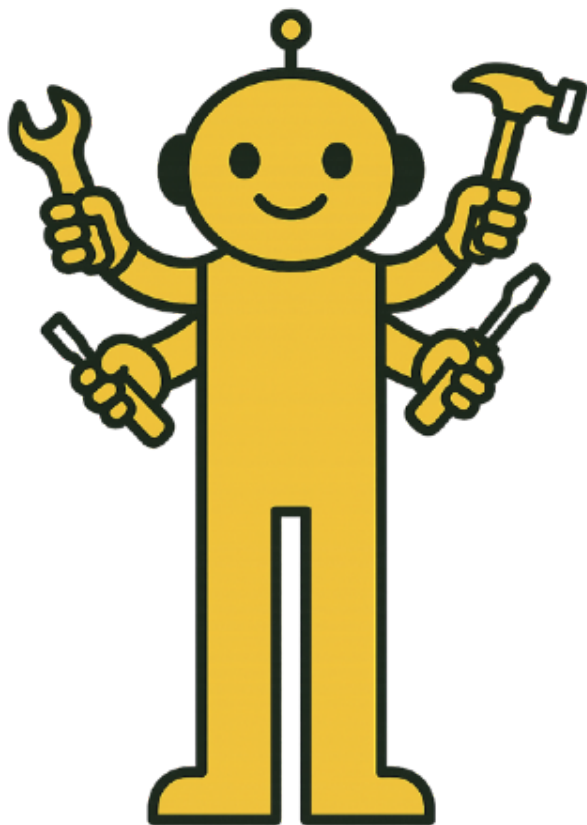


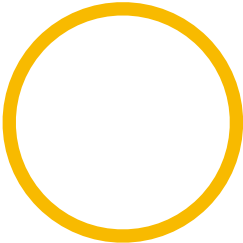






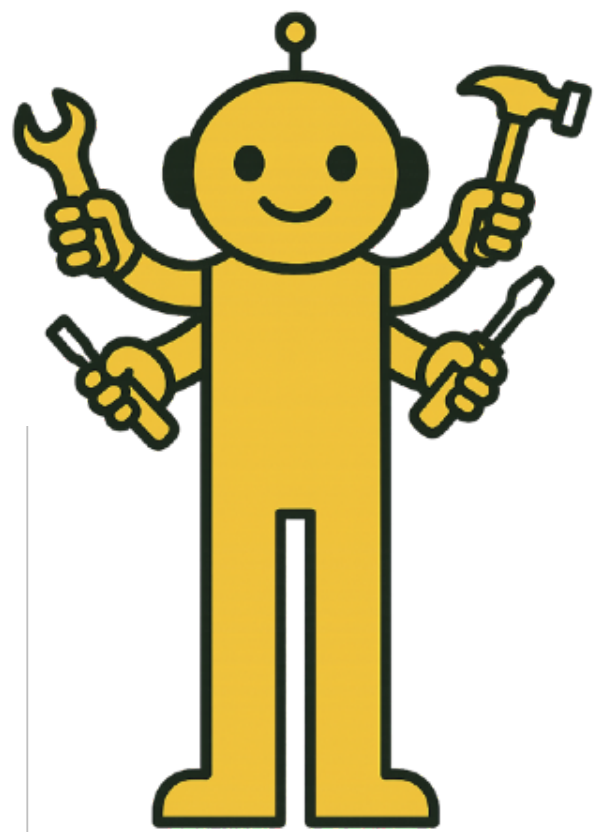
Iterators





Iterator agents

LLM **owns a loop**: plan → act → observe → repeat until done.

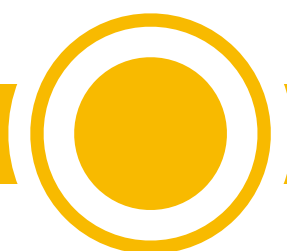


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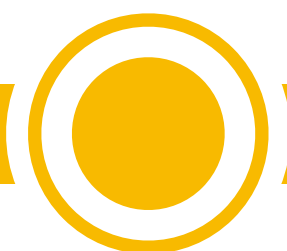
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Chat LLMs



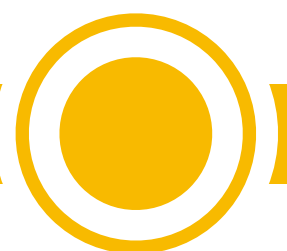
Routers



Tool callers



Iterators



Full Autonomy

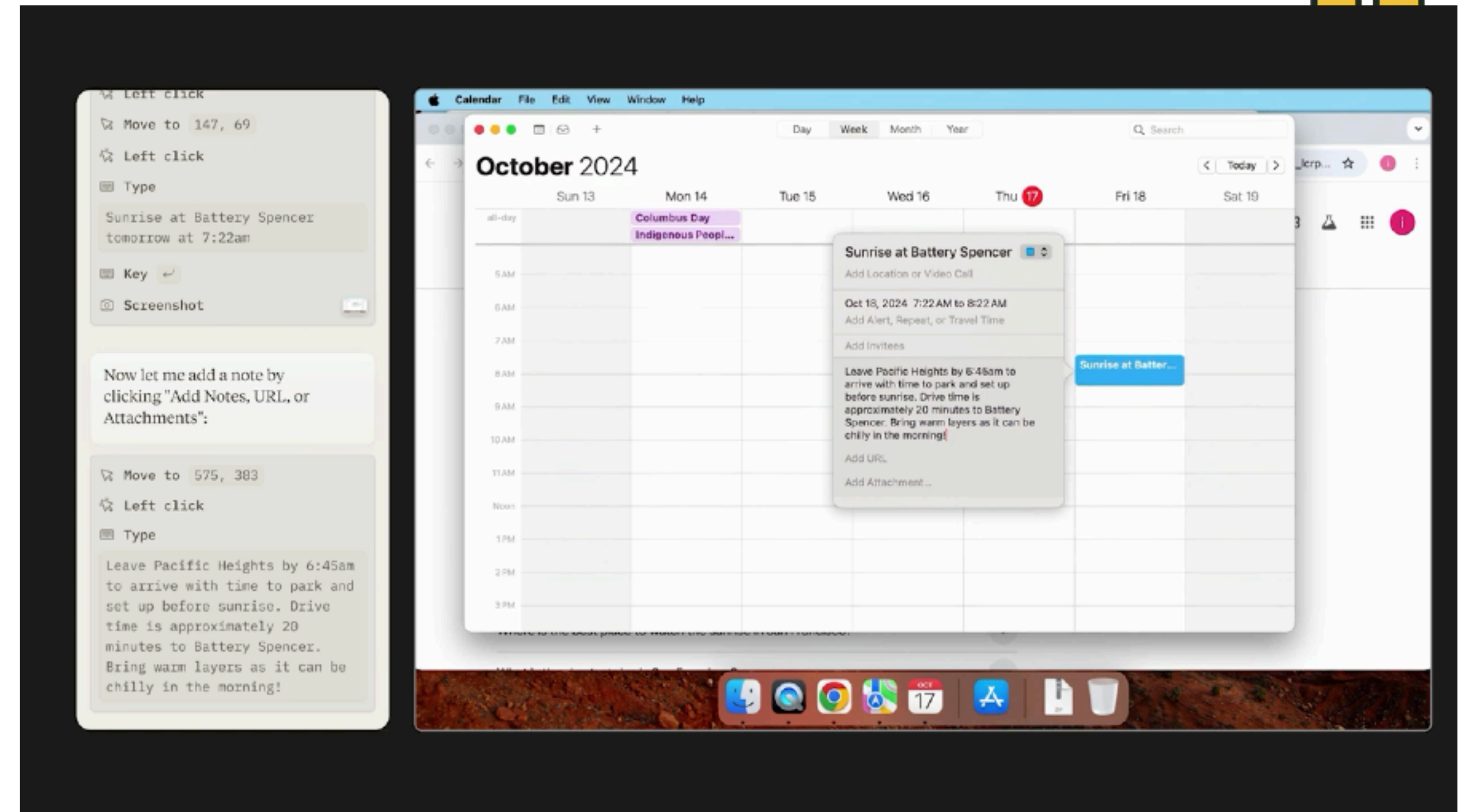
Iterator agents

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```
54 54 /// A transport implementation
55 55 #[derive(Clone)]
56 56 pub struct Transport {
57 57     addr: SocketAddr,
58 58     tls_enabled: bool,
59 59 }
60 60
61 61 impl Transport {
62 62     pub async fn new(config: TransportConfig) -> Result<Self, Box<dyn std::error::Er
63 63     {
64 64         Ok(Self {
65 65             addr: config.addr,
66 66             tls_enabled: config.tls_config.is_some(),
67 67         })
68 68     }
69 69
70 70     pub fn local_addr(&self) -> Result<SocketAddr, Box<dyn std::error::Error>> {
71 71         Ok(self.addr)
72 72     }
73 73
74 74     pub async fn accept(&self) -> Result<Stream, Box<dyn std::error::Error>> {
75 75         Ok(Stream::new())
76 76     }
77 77
78 78 /// A transport stream
79 79 pub struct Stream {}
80 80
81 81 impl Stream {
82 82     pub(crate) fn new() -> Self {
83 83         Self {}
84 84     }
85 85
86 86     pub async fn read(&mut self, _buf: &mut [u8]) -> std::io::Result<usize> {
87 87         Ok(0)
88 88     }
89 89 }
```

Cursor



Claude Computer Use

Other examples: GitHub Copilot, Amazon Q Developer agents, Windsurf agents, Zapier agents

Chat LLMs

Routers

Tool callers

Iterators

Full Autonomy